MACHINE LEARNING

MASTER SCIENCES DES DONNÉES ET ANALYTIQUE

FPS-ESTS

Based on the book of Goodfellow-et-al-2016

MAIN CONTENT:

- Machine learning involves developing models that generalize from data.
- **Key concepts:**
  - **Learning algorithms: target, performance measure P, experience E, an example (Linear Regression…)**
  - Capacity, overfitting, and underfitting
  - Hyperparameters and validation sets
  - Estimators, bias, and variance
  - Maximum likelihood estimation
  - Bayesian statistics
  - Supervised learning algorithms
  - Unsupervised learning algorithms
  - Stochastic gradient descent
  - Building a machine learning algorithm
  - Challenges driving deep learning

CHAPTER 1: LEARNING ALGORITHMS

- **What is a Learning Algorithm?**
  - A learning algorithm is a method used to automatically identify patterns in data and make predictions.
  - It learns from existing data and improves its performance over time.
  - **Example**: The linear regression algorithm, which models relationships between variables by fitting a straight line to data points.
  - Learning algorithms are the foundation of artificial intelligence and data-driven decision-making.

CHAPTER 1: LEARNING ALGORITHMS

- **Mitchell's Definition of Machine Learning**
  - **Definition by Mitchell (1997):** "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E."
  - **Key components:**
    - ✓ **Tasks (T):** The specific problems the algorithm aims to solve (e.g., image classification, spam detection).
    - ✓ **Experience (E):** The data the algorithm learns from (e.g., labeled training datasets).
    - ✓ **Performance Measure (P):** The metric used to evaluate improvement (e.g., accuracy, precision, recall).
- **Example:** A spam filter (task T) improves its accuracy (P) by analyzing more emails (E).

Mitchell, T. M. (1997). Machine Learning. McGraw-Hill, New York. 97

CHAPTER 1: LEARNING ALGORITHMS - TASKS

- Some of the most common machine learning **tasks** include :
  - **Classification,**
  - **Classification with missing inputs,**
  - **Regression,**
  - **Transcription,**
  - **Machine translation,**
  - **Structured output,**
  - **Anomaly detection,**
  - **Synthesis and sampling,**
  - **Imputation of missing values,**
  - **Denoising,**
  - **Density estimation or probability mass function estimation**
  - **…**

CHAPTER 1: LEARNING ALGORITHMS - TASKS

- **Classification in Machine Learning**

  - **Definition** : Classification is a type of supervised machine learning task where the goal is to predict which of several categories an input belongs to.
  - **Function** :The learning algorithm learns a function $f; \mathbb{R}^n \rightarrow \{1, \ldots k\}$ which maps an input vector x to a class label y, where y is a numeric code representing one of the possible categories
  - **Example** : Object recognition — identifying objects from images.
    - A typical example of classification is object recognition, where the input is an image and the output is a class label (e.g., identifying whether the image contains a cat, dog, or car

CHAPTER 1: LEARNING ALGORITHMS - TASKS

- **Classification in Machine Learning**
  - **Example** of Object recognition — identifying objects from images.
    - For instance, the Willow Garage PR2 robot can recognize different drinks (e.g., coffee, juice, etc.) and serve them to users upon request (Goodfellow et al., 2010). The robot uses image data from its cameras, which are processed to identify the type of drink.
    - **Application** : This capability is used in robotics for tasks such as serving drinks, and is also foundational in systems that enable automatic tagging of objects in photos or videos.



CHAPTER 1: LEARNING ALGORITHMS - TASKS

- **Classification with Missing Inputs**
  - **Definition:** Classification becomes more challenging when a program is not guaranteed that every measurement in its input vector will always be provided.
  - **Problem Statement:** Traditional classification models assume all features are available, but in real-world applications, missing inputs are common.
  - **Why Inputs May Be Missing**
    - **Common Causes:**
      - Data corruption or loss
      - Privacy concerns and restricted data access
      - Expensive or invasive medical tests
      - Sensor failures in automated systems
  - **Example:** Medical diagnosis where certain tests are not performed on all patients

CHAPTER 1: LEARNING ALGORITHMS - TASKS

- **Classification with Missing Inputs**
  - **Standard Approach:** Learning a single function mapping input vectors to categorical outputs.
  - **Challenge:** When inputs are missing, the model must learn multiple classification functions depending on available features.
  - Instead of one function, the algorithm must learn **a set of functions**, each corresponding to different missing input subsets.
  - If there are **n** input variables, up to $\mathbf{2^n}$ different classification functions may be needed.
  - **Example:** A model trained on complete data but tested with incomplete patient records
  - **Solution:** Learn a **probability distribution** over all relevant variables.
  - **Method:** Marginalize out missing variables to infer the most probable classification.
  - **Key Benefit:** Instead of learning 2^n different functions, the model only needs to learn **one probability function**

CHAPTER 1: LEARNING ALGORITHMS - TASKS

- **Regression in Machine Learning**
  - **Definition:** Regression involves predicting a numerical value given some input.
  - **Mathematical Representation:** The learning algorithm outputs a function f: $R^n \rightarrow R$.
  - **Key Difference from Classification:** The output is a continuous value rather than a categorical label
  - **How Regression Works**
    - **Input:** A set of features $(x_1, x_2, ..., x_n)$
    - **Output :** A continuous numerical value y
    - **Example:** Predicting house prices based on square footage, location, and amenities (équipements)

CHAPTER 1: LEARNING ALGORITHMS - TASKS

- **Regression in Machine Learning**
  - **Applications of Regression**
    - **Insurance Premium Prediction:** Estimating expected claim amounts for policyholders (Estimation des montants de sinistres attendus pour les assurés).
    - **Stock Market Forecasting:** Predicting future prices of securities for algorithmic trading (refers to using computer programs and algorithms to execute financial market trades at high speed and frequency).
    - **Weather Forecasting:** Estimating future temperature, precipitation levels, etc.
    - **Healthcare Analytics:** Predicting patient recovery time or disease progression

CHAPTER 1: LEARNING ALGORITHMS - TASKS

- **Transcription in Machine Learning**
  - **Definition:** Machine learning systems transcribe unstructured data into discrete textual form.
  - **Examples:** Optical character recognition (OCR) and speech recognition.
  - **Real-World Usage:** Google Street View address processing, speech-to-text applications:
    - **Optical Character Recognition (OCR)**
      - **Task:** Convert text in images to digital characters.
      - **Example:** Identifying house numbers from Google Street View images.
      - **Technology:** Deep learning models detect and classify characters in an image
    - **Speech Recognition**
      - **Task:** Convert spoken language into textual form.
      - **Input:** Audio waveform.
      - **Output:** Sequence of characters or word IDs.
      - **Example:** Transcribing a conversation for real-time captions.
    - **Deep Learning in Transcription (RNNs, CNNs)** used at major companies, including Microsoft, IBM and Google

CHAPTER 1: LEARNING ALGORITHMS - TASKS

- **Machine Translation in Machine Learning**
  - **Definition:** Machine learning systems translate text from one language to another.
  - **Examples:** Translating English to French, Spanish to Chinese.
  - **Real-World Usage:** Google Translate, Microsoft Translator, AI-powered language tools
  - **How Machine Translation Works:**
    - **Input:** A sequence of symbols in the source language.
    - **Output:** A corresponding sequence of symbols in the target language.
    - **Example:** Translating "Hello, how are you?" into "Bonjour, comment ça va?«
  - **Deep learning has recently begun to have an important impact on this kind of task**
    - Captures contextual meaning.
    - Learns from large multilingual datasets

CHAPTER 1: LEARNING ALGORITHMS - TASKS

- **Structured Output in Machine Learning**
  - **Definition:** Tasks where the output is a vector or data structure with interrelated values.
  - **Examples:** Parsing, image segmentation, and image captioning.
  - **Why It Matters:** Many real-world problems require structured relationships in outputs
    - **Parsing as a Structured Output Task**
      - **Task:** Mapping a natural language sentence into a syntactic tree.
      - **Example:** Identifying nouns, verbs, and adverbs in a sentence.
      - **Deep Learning Application:** Used for parsing sentence structures (Collobert, 2011)
    - **Image Segmentation**
      - **Task:** Assigning each pixel in an image to a specific category.
      - **Example:** Identifying roads in aerial photographs.
      - **Deep Learning Application:** Mnih and Hinton (2010) applied deep learning to aerial image analysis.
    - **Image Captioning**
      - **Task:** Generating a natural language description for an image.
      - **Example:** AI describing an image as "A cat sitting on a sofa."
      - **Deep Learning Models:**
        - Recurrent Neural Networks (RNNs)
        - Convolutional Neural Networks (CNNs)
        - Attention Mechanisms (Xu et al., 2015)

CHAPTER 1: LEARNING ALGORITHMS - TASKS

- **Anomaly Detection in Machine Learning**
  - **Definition:** Detecting unusual or atypical data points in a dataset.
  - **Importance:** Used to identify fraud, system failures, and security threats.
  - **Examples:** Credit card fraud detection, network intrusion detection.
  - **How Anomaly Detection Works**
    - **Step 1:** Collect and preprocess data.
    - **Step 2:** Build a model to learn normal behavior.
    - **Step 3:** Detect deviations from the learned patterns.
    - **Step 4:** Flag and analyze anomalies

CHAPTER 1: LEARNING ALGORITHMS - TASKS

- **Anomaly Detection in Machine Learning**
  - **Credit Card Fraud Detection**
    - **Task:** Identify fraudulent transactions based on user purchasing behavior.
    - **Method:** Model normal spending patterns and flag unusual transactions.
    - **Example:** If a purchase is made in a foreign country suddenly, it may be flagged as fraud
  - **Machine Learning Approaches for Anomaly Detection**
    - **Supervised Learning:** Requires labeled normal and anomalous data.
    - **Unsupervised Learning:** Detects anomalies without labeled examples.
    - **Semi-Supervised Learning:** Uses mostly normal data with a few labeled anomalies
  - **Common Anomaly Detection Algorithms**
    - **Statistical Methods:** Z-score, Gaussian Mixture Models.
    - **Machine Learning Models:**
      - Isolation Forest
      - One-Class SVM
      - Autoencoders (Deep Learning)

- **Applications of Anomaly Detection**
  - **Financial Sector:** Fraud detection in credit card transactions.
  - **Healthcare**: Identifying rare diseases in patient records.
  - **Cybersecurity:** Detecting network intrusions.
  - **Manufacturing:** Identifying defective products in assembly lines

CHAPTER 1: LEARNING ALGORITHMS - TASKS

- **Synthesis and Sampling in Machine Learning**
  - **Definition: Machine learning models generate new examples similar to training data.**
  - **Importance: Enables large-scale content generation for media, gaming, and automation.**
  - **Examples: AI-generated textures, speech synthesis, image generation Common Anomaly Detection Algorithms**
  - **How Synthesis and Sampling Work:**
    - **Step 1: Train a model on an existing dataset.**
    - **Step 2: Generate new data points by sampling from the learned distribution.**
    - **Step 3: Refine outputs to ensure diversity and realism**
  - **Applications of Synthesis and Sampling**
    - **Gaming: AI-driven environment creation.**
    - **Healthcare: AI-generated synthetic medical data for privacy.**
    - **Entertainment: AI-powered music and art generation.**
    - **Cybersecurity: Generating synthetic data for adversarial training.**

CHAPTER 1: LEARNING ALGORITHMS - TASKS

- **Imputation of Missing Values in Machine Learning**
  - **Definition:** Machine learning models predict missing values in a dataset.
  - **Importance:** Ensures data completeness for accurate model performance.
  - **Examples:** Filling in missing survey responses, reconstructing medical records
  - **How Imputation works :**
    - **Step 1:** Identify missing data points in a dataset.
    - **Step 2:** Use statistical or machine learning models to estimate missing values.
    - **Step 3:** Integrate imputed values into the dataset for analysis

- **Applications of Missing Data Imputation**
  - **Healthcare: Filling in incomplete patient records.**
  - **Finance: Completing missing transaction details.**
  - **Survey Analysis: Predicting missing responses for better insights.**
  - **Cybersecurity: Reconstructing incomplete threat detection logs**

CHAPTER 1: LEARNING ALGORITHMS - TASKS

- **Denoising in Machine Learning**
  - **Definition:** Predicting clean data from a corrupted version using machine learning.
  - **Importance:** Enhances data quality for analysis and model performance.
  - **Examples:** Image denoising, audio enhancement, text correction
  - **How Imputation works :**
    - **Step 1:** Identify and quantify noise in input data.
    - **Step 2:** Train models to map corrupted data to its clean version.
    - **Step 3:** Output a denoised version or a probability distribution of possible clean data
  - **Types of Noise in Data**
    - **Gaussian Noise:** Common in images and sensor data.
    - **Salt-and-Pepper Noise :** Appears as random black-and-white pixels
    - **Missing Data Noise:** Occurs in incomplete datasets.
    - **Compression Artifacts:** Loss of quality due to excessive data compression
  - **Applications of Denoising :**
    - **Image Processing:** Removing noise from photos and medical images.
    - **Speech and Audio Enhancement:** Improving voice clarity in noisy environments
    - **Text Correction:** Detecting and fixing OCR or speech-to-text errors.

CHAPTER 1: LEARNING ALGORITHMS -THE PERFORMANCE MEASURE, P

- **Introduction to Performance Evaluation**
  - **Definition:** Measuring how well a machine learning algorithm performs a given task.
  - **Importance:** Determines real-world effectiveness of ML models.
  - **Examples: :** Classification accuracy, regression error, probabilistic model evaluation.
  - **Choosing a Performance Measure**
    - **Task-Specific Metrics:** Different ML tasks require different evaluation criteria.
    - **Example:** Classification models use accuracy/error rate; density estimation uses log-probability.
    - **Objective:** Ensure the measure aligns with the intended system behavior

CHAPTER 1: LEARNING ALGORITHMS -THE PERFORMANCE MEASURE, P

- **Classification Performance Metrics**
  - **Accuracy:** Proportion of correctly classified examples.
  - **Error Rate:** Proportion of misclassified examples.
  - **0-1 Loss:**
    - **0:** Correct classification
    - **1:** Incorrect classification
- **Regression Performance Metrics**
  - **Mean Absolute Error (MAE):** Measures average prediction error.
  - **Mean Squared Error (MSE):** Penalizes larger errors more than smaller ones.
  - **Trade-Off:** Consider whether frequent small errors or rare large errors are more important

CHAPTER 1: LEARNING ALGORITHMS -THE PERFORMANCE MEASURE, P

- **Density Estimation Performance**
  - **Why Accuracy Doesn't Apply:** No clear correct/incorrect output. Performance metric should give the model a continuous-valued score for each example
  - **Log-Probability:** Measures how well the model estimates data distribution.
  - **Challenges:** Some probabilistic models only implicitly represent probability distributions gives the model a continuous-valued score for each example
  - **Density estimation** is used when we need to understand the probability distribution of data, especially in probabilistic models and generative tasks:
    - **Anomaly Detection:** Identifies outliers by checking how probable a data point is.
    - **Generative Modeling:** Used in algorithms like VAEs and GANs to generate new samples
    - **Probability Estimation:** Helps in decision-making where knowing uncertainty is crucial

CHAPTER 1: LEARNING ALGORITHMS -THE PERFORMANCE MEASURE, P

- **The choice of performance measure by Evaluating the model on Unseen Data**
  - **Why It Matters:** Generalization determines real-world effectiveness.
  - **Method:** Use a separate test set for evaluation.
  - **Risk of Overfitting:** Models may perform well on training data but fail on new data.
- **Challenges in Choosing a Performance Measure**
  - **Complexity in Defining Success:**
    - Transcription: Entire sequence vs. partial correctness (should we measure the accuracy of the system at transcribing entire sequences, or should we use a more fine-grained performance measure that gives partial credit for getting some elements of the sequence correct?)
    - Regression: Medium vs. rare large errors(should we penalize the system more if it frequently makes medium-sized mistakes or if it rarely makes very large mistakes?)
  - **Measurement Feasibility:** Some ideal metrics are impractical to compute

CHAPTER 1: LEARNING ALGORITHMS -THE EXPERIENCE

- **Understanding Learning Through Experience**
  - Machine learning algorithms can be broadly categorized as supervised or unsupervised.
  - Categorization depends on the type of **experience** allowed during the learning process
- **Learning Through Experience**
  - Most algorithms learn from an entire dataset.
  - A dataset consists of multiple examples, also referred to as data points

CHAPTER 1: LEARNING ALGORITHMS -THE EXPERIENCE

- **Example Dataset – The Iris Dataset**
  - One of the oldest datasets used in machine learning research.
  - Introduced by Fisher in 1936.
  - Consists of 150 iris plants with measurements of:
    - Sepal length
    - Sepal width
    - Petal length
    - Petal width
  - Each example is labeled with one of three species



CHAPTER 1: LEARNING ALGORITHMS -THE EXPERIENCE

- **Understanding Supervised Learning**
  - Supervised learning uses labeled data.
  - The algorithm learns to map inputs to outputs.
  - Example: Classifying iris plants into species based on their measurements.
  - Supervised learning algorithms **experience** a dataset containing features, but each example is also associated with a label or target.
  - Example: The Iris dataset is annotated with the species of each iris plant.
  - A supervised learning algorithm can study the dataset and classify iris plants into three different species based on measurements.
  - The term "supervised learning" comes from the idea that a target (label) is provided by an instructor or teacher, guiding the learning process

CHAPTER 1: LEARNING ALGORITHMS -THE EXPERIENCE

- **Understanding Unsupervised Learning**
  - Unsupervised learning algorithms analyze a dataset with many features.
  - They learn patterns and structures without labeled outputs
  - Unlike supervised learning, there is no instructor or teacher providing labels.
  - Unsupervised learning involves observing multiple examples of a random vector x and attempting to learn the probability distribution $p(x)$ or interesting properties of that distribution.
- **Unsupervised Learning Applications**
  - **Density Estimation:** Learning the probability distribution of data.
  - **Clustering:** Grouping data into clusters of similar examples.
  - **Denoising & Synthesis:** Enhancing and generating new data

CHAPTER 1: LEARNING ALGORITHMS -THE EXPERIENCE

- **Bridging Supervised and Unsupervised Learning**
  - Unsupervised learning and supervised learning are not formally defined terms, and the boundaries between them are often blurred.
  - Many machine learning technologies can be applied to both tasks.
  - The **chain rule of probability** states that for a vector $x \in \mathbb{R}^n$, the joint distribution can be decomposed as:

$$p(x) = \prod_{i=1}^{n} p(x_i | x_1, \ldots, x_{i-1})$$

This means that an unsupervised problem of modeling $p(x)$ can be solved by splitting it into n supervised learning problems.

- Example: Suppose you are drawing two cards from a deck without replacement:
  - $x_1 = drawing\ an\ Ace\ first, x_2 = drawing\ a\ King\ second,$
  - Using the chain rule: $P(x_1, x_2) = P(x_1)P(x_2|x_1)$
  - $P(x_1) = \frac{4}{52}$, (since there are 4 Aces in a 52-card deck)
  - $P(x_2|x_1) = \frac{4}{51}$, (since one card has already been removed). Thus
  - $P(x_1, x_2) = \frac{4}{52} \times \frac{4}{51} = \frac{16}{2652} \approx 0{,}00603$

CHAPTER 1: LEARNING ALGORITHMS -THE EXPERIENCE

- **Bridging Supervised and Unsupervised Learning**
- Alternatively, supervised learning of $p(y \mid x)$ can leverage unsupervised learning techniques by first modeling the joint distribution $p(x, y)$ and then inferring:

$$p(y|x) = \frac{p(x|y)}{\sum_{y'} p(x|y')}$$

- This is a **normalized likelihood function**, which follows from **Bayes' Theorem** in probabilistic learning.
- $p(y|x) \rightarrow$ **Posterior probability**: The probability of label y given an input x.
- $p(x|y) \rightarrow$ **Likelihood**: The probability of observing x given class y.
- $\sum_{y'} p(x|y') \rightarrow$ **Normalization factor**: Ensures that the probabilities sum to 1.
- **Interpretation**
  - Instead of directly estimating $p(y|x)$ (as in traditional supervised learning), we estimate the likelihood $p(x|y)$ from an **unsupervised** generative model.
  - Once we have $p(x|y)$, we compute $p(y|x)$ by normalizing over all possible labels y′.
  - This approach is common in **generative models** like Naïve Bayes, Hidden Markov Models, and Gaussian Mixture Models (Applications: Image and Video Generation, Text Generation…).

CHAPTER 1: LEARNING ALGORITHMS -CONCLUSION

- **Learning Paradigms:**
  - Machine learning is broadly categorized into supervised and unsupervised learning.
  - These categories are not strictly formal but help in understanding different approaches.
  - Other learning paradigms also exist, including semi-supervised learning, multi-instance learning, and reinforcement learning.

CHAPTER 1: LEARNING ALGORITHMS -CONCLUSION

- **Supervised Learning**
  - Involves learning a function that maps inputs to outputs based on labeled examples.
  - Common tasks:
    - Regression: Predicting continuous values.
    - Classification: Categorizing inputs into discrete labels.
    - Structured Output Problems: Generating structured data like translations or images.

- **Unsupervised Learning**
  - Involves learning patterns and structures in data without explicit labels.
  - Common tasks:
    - Clustering: Grouping similar data points.
    - Density Estimation: Estimating probability distributions.
    - Dimensionality Reduction: Reducing data complexity while preserving structure.

CHAPTER 1: LEARNING ALGORITHMS -CONCLUSION

- **Other Learning Paradigms**
  - **Semi-Supervised Learning:**
    - Combines labeled and unlabeled data.
    - The model is trained on a small amount of labeled data and a large amount of unlabeled data.
    - Useful when labeling data is expensive or time-consuming, but unlabeled data is abundant.
    - **Example1: Google Photos Auto-Tagging**
      - A small subset of images is manually labeled (e.g., "dog," "cat," "car").
      - The model learns patterns from labeled images and applies them to the vast pool of unlabeled images.
      - Used in facial recognition and object identification.
    - **Example2: Automatic Speech Transcription**
      - A small amount of transcribed speech data (labeled) is available.
      - The system learns from a vast collection of spoken language without transcripts (unlabeled).
      - Used in virtual assistants like **Siri, Google Assistant, and Alexa**.

CHAPTER 1: LEARNING ALGORITHMS -CONCLUSION

- **Other Learning Paradigms**
  - **Multi-Instance Learning:** Labels are assigned to collections of instances rather than individual instances. The goal is to determine how the presence of certain instances influences the bag's label.
  - **Example1: Predicting Drug Effectiveness**
    - Each molecule (bag) contains multiple conformations (instances).
    - The molecule is labeled as "active" if at least one conformation binds to a target protein.
    - **Use Case:** Drug discovery and virtual screening in bioinformatics.
  - **Example2: Identifying Cancer in Whole Slide Images (WSI)**
  - A **histopathology slide** (bag) contains thousands of small patches (instances).
  - If at least one patch contains cancerous tissue, the whole slide is labeled "cancerous."
  - **Use Case:** AI-powered diagnosis in medical imaging (e.g., detecting tumors from biopsy scans).

  - **Reinforcement Learning:** The model interacts with an environment and learns via feedback loops.

CHAPTER 1: LEARNING ALGORITHMS -CONCLUSION

- **Other Learning Paradigms**
  - **Reinforcement Learning:** Unlike traditional machine learning , the model interacts with an environment and learns via feedback loops.
  - Examples:
    - Training agents in video games (Deep Q-Networks).
    - Robotics and autonomous systems.
  - Further reading: Sutton & Barto (1998), Bertsekas & Tsitsiklis (1996)

CHAPTER 1: LEARNING ALGORITHMS -CONCLUSION

- **Understanding Datasets**
  - Most machine learning algorithms operate on datasets.
  - A dataset is a collection of examples, each containing features.
  - Example: The **Iris dataset** has 150 examples, each with 4 features.
    - Can be represented as a design matrix: $X \in \mathbb{R}^{150\times4}$.
- **Design Matrices in Machine Learning**
  - A **design matrix** represents a dataset as a structured table:
    - Each row: A different example.
    - Each column: A different feature.
  - Example: **Iris Dataset**
    - $X_{i,1}$= Sepal length of plant *i*.
    - $X_{i,2}$ = Sepal width of plant *i*, etc.

CHAPTER 1: LEARNING ALGORITHMS -CONCLUSION

- **Handling Heterogeneous Data**
  - Not all datasets can be represented as a design matrix.
  - **Example:**
  - Images of different sizes have varying pixel counts.
  - Instead of a matrix, use a set notation: $\{x^{(1)}, x^{(2)}, \ldots, x^{(m)}\}$. This notation does not imply that any two example vectors $x^{(i)}, x^{(j)}$ have the same size
- **Conclusion**
  - Machine learning encompasses multiple paradigms, from supervised to reinforcement learning.
  - Understanding dataset structures is essential for applying learning algorithms.
  - Continuous innovation allows for new learning methods tailored to unique problems.

CHAPTER 1: LEARNING ALGORITHMS -EXAMPLE

- **Example: Linear Regression**
  - Machine learning improves a computer program's performance at a task via experience.
  - Linear regression is a simple machine learning algorithm that solves a regression problem.
  - The goal is to predict a scalar output $y \in \mathbb{R}$ from an input vector $\mathrm{x} \in \mathbb{R}^n$.
  - The output is modeled as a linear function:

$$\hat{y} = w^T x, where\ w \in \mathbb{R}^n\ is\ a\ vector\ of\ parameters$$

  - **Interpreting Parameters:**
    - Each weight w_i controls how feature x_i influences the prediction.
    - **Positive weight:** Increasing x_i increases $\hat{y}$.
    - **Negative weight:** Increasing x_i decreases $\hat{y}$.
    - **Zero weight:** The feature has no effect on $\hat{y}$.
    - **Large weight magnitude:** The feature has a strong influence on $\hat{y}$.

CHAPTER 1: LEARNING ALGORITHMS -EXAMPLE

- **Example: Linear Regression**
  - Next, we need a definition of our performance measure, P.
  - One way of measuring the performance of the model is to compute the **Mean Squared Error (MSE)**:

$$MSE_{test} = \frac{1}{m}\sum_{i}(\hat{y}^{(test)} - y^{(test)})_i^2$$

  - $\hat{y}^{(test)}$ gives the predictions of the model on the test set
  - $y^{(test)}$ is the vector of regression targets
  - MSE_{test} decreases to 0 when $\hat{y}^{(test)} = y^{(test)}$
  - We can also see that: $MSE_{test} = \frac{1}{m}\left\|\hat{y}^{(test)} - y^{(test)}\right\|_2^2$
  - So the error increases whenever the Euclidean distance between the predictions and the targets increases

CHAPTER 1: LEARNING ALGORITHMS -EXAMPLE

- **Example: Linear Regression**
- To minimize MSE_{train}, we can simply solve for where its gradient is 0

$$\nabla MSE_{train} = 0 \Rightarrow \nabla_w \frac{1}{m} \left\| X^{(train)} w - y^{(train)} \right\|_2^2 = 0$$

  - Expanding: $\nabla_w (X^{(train)} w - y^{(train)})^T (X^{(train)} w - y^{(train)}) = 0$
  - Applying the gradient: $\nabla_w (w^T X^{(train)T} X^{(train)} w - 2 w^T X^{(train)T} y^{(train)} + y^{(train)T} y^{(train)}) = 0$
  - $\Rightarrow 2 X^{(train)T} X^{(train)} w - 2\, X^{(train)T} y^{(train)} = 0$
  - $\Rightarrow \mathrm{w} = (X^{(train))T} X^{(train)})^{-1} X^{(train)T} y^{(train)}$
- This is known as the normal equations
- Evaluating this equation constitutes a simple learning algorithm

CHAPTER 1: LEARNING ALGORITHMS -EXAMPLE

- **Example: Linear Regression with Intercept Term**
  - The term **linear regression** is often used to refer to a more sophisticated model with an **intercept term** .
  - This model is given by: $\hat{y} = w^T x + b$
  - The mapping from parameters to predictions remains **linear**, but the mapping from **features to predictions** is now an **affine function**.
  - The **bias parameter** b shifts the predictions away from the origin.
  - This extension allows linear regression to model relationships that do not necessarily pass through the origin.
  - The term **"linear"** is often used to describe affine functions in machine learning.
  - Linear regression, though simple, serves as a foundation for more advanced learning algorithms.

CHAPTER 1: LEARNING ALGORITHMS -EXAMPLE

- **Example: Linear Regression with Intercept Term** (Goodfellow-et-al-2016)

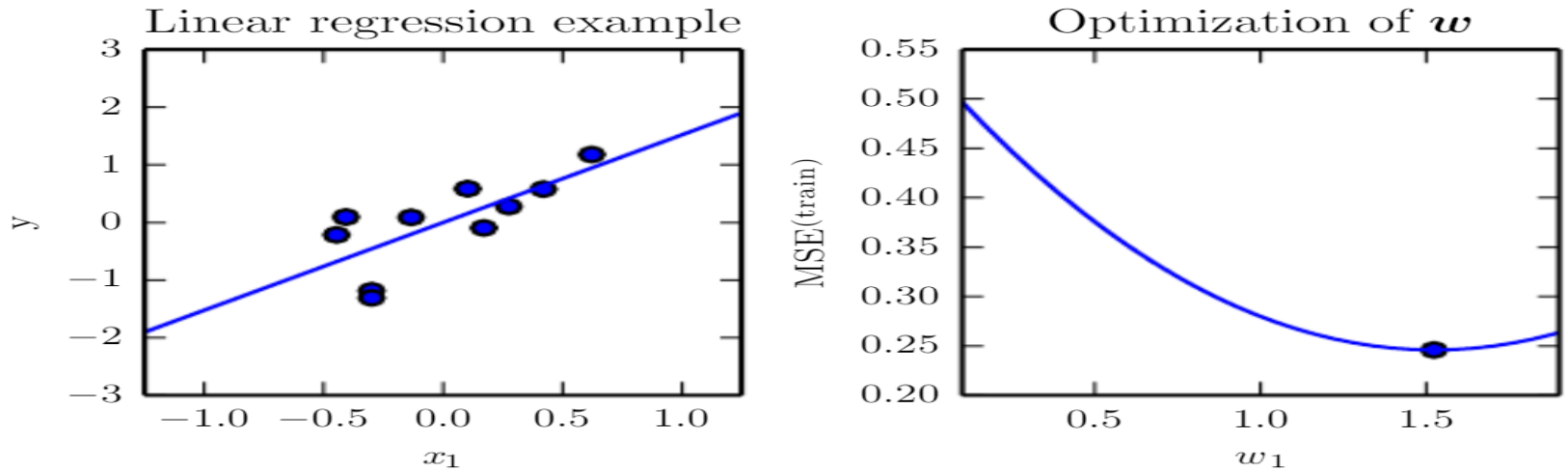


Figure 5.1: A linear regression problem, with a training set consisting of ten data points, each containing one feature. Because there is only one feature, the weight vector $\boldsymbol{w}$ contains only a single parameter to learn, w_1. *(Left)*Observe that linear regression learns to set w_1 such that the line $y = w_1 x$ comes as close as possible to passing through all the training points. *(Right)*The plotted point indicates the value of w_1 found by the normal equations, which we can see minimizes the mean squared error on the training set.